

Program : Diploma in Artificial Intelligence and Machine Learning	
Course Code : 6349B	Course Title: Robotic Process Automation Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide hands on experiments with the basic concepts of Robotic Process Automation
- To learn how to install UiPath Studio Community Edition.
- To Develop Bots using UiPath.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Sound Knowledge of basic programming concepts	2139	Problem Solving and Programming Lab	II

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Installation and Familiarization of UiPath Studio Community Edition tool	7	Applying
CO2	Develop Bots using types of variables and arguments	8	Applying
CO3	Implement Bots using control flow statements	10	Applying
CO4	Develop Bots using Basic and Desktop recording	17	Applying
	Series Test	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3 – Strongly mapped

2 – Moderately mapped

1 – Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Installation and Familiarization of UiPath Studio Community Edition tool		
M1.01	Installation of UiPath Studio Community Edition	1	Applying
M1.02	Familiarization of UiPath components- Activities and Sequence	2	Applying
M1.03	Familiarization of the UiPath Interface components- The Ribbon, The Universal Search Bar, The Designer Panel	2	Applying
M1.04	Familiarization of the UiPath Interface components- The Activities Panel, The Library Panel, The Project Panel, The Properties Panel, The Outline Panel	2	Applying

CO2	Develop Bots using types of variables and arguments		
M2.01	Create a simple bot which will greet you with the name i.e. “Welcome to UiPath Studio”	1	Applying
M2.02	Create a welcome message using variable	1	Applying
M2.03	Develop an automation using text variables	1	Applying
M2.04	Implement an automation using True or False variables.	1	Applying
M2.05	Develop an automation using integer variables	1	Applying
M2.06	Implement an automation using array of variables	1	Applying
M2.07	Create an automation using date and time variable	1	Applying
M2.08	Develop an automation using arguments	1	Applying
	Series Test 1	1.5	
CO3	Implement Bots using control flow statements.		
M3.01	Create an automation using delay activity	2	Applying
M3.02	Develop a bot using if activity – Example: build a bot that prints a message based on whether a cell has a value or is empty	2	Applying
M3.03	Develop a bot using switch activity – Build a bot which asks the user to input some numbers and perform simple arithmetic operations like addition, subtraction, multiplication and division as per users wish and displays the result as an output message.	2	Applying
M3.04	Implement an automation using do-while, while activity	2	Applying
M3.05	Create a bot using for each activity	2	Applying
CO4	Develop Bots using Basic and Desktop recording.		
M4.01	Create a simple bot to automate a manual task which includes a sequence of mouse movements and keyboards	2	Applying
M4.02	Implement a simple bot to automate a manual task which opening a desktop application and perform some activity like opening notepad	2	Applying

	and typing some data into it and saving the text file.		
M4.03	Create a simple bot to open a browser and visit a website/url of interest and then close the browser	2	Applying
M4.04	Develop a simple bot to read the contents of an excel sheet and print them in a message box.	3	Applying
M4.05	Create Bots that performs simple excel data manipulation	3	Applying
M4.06	Create a Screen scrapping Bot	5	Applying
	Series Test 2	1.5	

**** - Open-ended Experiment**

Students are expected to do a micro project using UiPath Studio edition tool during the course for continuous evaluation. .

Example:

A simple screen scrapping robot

Excel data sorting etc.

Text / Reference:

T/R	Book Title/Author
T1	Alok Mani Tripathi, “Learning Robotic Process Automation”, Packt Publishing, 2018
R1	Frank Casale , Rebecca Dilla, Heidi Jaynes , Lauren Livingston, “Introduction to Robotic Process Automation: a Primer”, Institute of Robotic Process Automation,1st Edition 2015.
R2	Richard Murdoch, Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant”, Independently Published, 1st Edition 2018.

Online Resources:

Sl.No	Website Link
1	https://docs.uipath.com/studio/docs

2	https://dev.tutorialspoint.com/uipath/uipath_tutorial.pdf
3	https://www.udemy.com/course/robotic-process-automation/